DiffFuSR: Super-Resolution of All Sentinel-2 Multispectral Bands Using Diffusion Models

Muhammad Sarmad, Michael Kampffmeyer, *Senior Member, IEEE*, and Arnt-Børre Salberg, *Member, IEEE*

***Abstract*—This article presents *DiffFuSR*, a modular pipeline for super-resolving all 12 spectral bands of Sentinel-2 Level-2A imagery to a unified ground sampling distance (GSD) of 2.5 m. The pipeline comprises two stages: 1) a diffusion-based super-resolution (SR) model trained on high-resolution (HR) RGB imagery from the NAIP and WorldStrat datasets, harmonized to simulate Sentinel-2 characteristics; and 2) a learned fusion network that upscales the remaining multispectral bands using the super-resolved RGB image as a spatial prior. We introduce a robust degradation model and contrastive degradation encoder to support blind SR. Extensive evaluations of the proposed SR pipeline on the OpenSR benchmark demonstrate that the proposed method outperforms current SOTA baselines in terms of reflectance fidelity, spectral consistency, spatial alignment, and hallucination suppression. Furthermore, the fusion network significantly outperforms classical and learnedpansharpening approaches, enabling accurate enhancement of Sentinel-2's 20- and 60-m bands. This work proposes a novel modular framework Sentinel-2 SR that utilizes harmonized learning with diffusion models and fusion strategies. Our code and models can be found at https://github.com/NorskRegnesentral/DiffFuSR**

***Index Terms*—Data fusion, diffusion models, Sentinel-2, super-resolution (SR), Wald protocol.**

I. Introduction

THE democratization of Earth observation has been significantly propelled by the advent of open-access satellite missions such as Sentinel-2, which offer freely available multispectral imagery on a global scale. These data serve as a valuable resource for a broad spectrum of applications, including land cover mapping, agricultural monitoring, urban planning, and disaster response [1]. There exist several other satellite sensors that provide RGB and multispectral imagery with higher resolution than Sentinel-2, e.g., WorldView 2 and 3 and Legion from Maxar, Pléiades from Airbus, and PlanetScope from Planet Labs. However, this data is not freely available, and the acquisition costs may be substantial. Given the global coverage and open availability of Sentinel-2's data, enhancing the spatial resolution through computational means is therefore a highly attractive prospect [2], [3], [4], [5], [6].

SR refers to reconstructing high-resolution (HR) images from their lower resolution counterparts, often using data-driven or model-based techniques. SR is an inherently ill-posed problem, as multiple HR images can produce the same LR counterpart, requiring models to infer lost high-frequency details. Moreover, SR is also criticized for its reliance on learned priors and the risk of generating hallucinated details [7]. However, when applied judiciously, with a clear understanding of its limitations, it can significantly improve both the visual interpretability and the utility of satellite imagery in downstream analytical tasks.

In the context of Sentinel-2, which consists of multispectral bands in 10-, 20-, and 60-m resolution, an opportunity arises to combine SR with data fusion techniques to enhance spatial resolution further. Unlike conventional SR, which relies solely on LR inputs, fusion approaches, such as pansharpening, leverage the higher resolution spectral bands (e.g., the 10-m RGB bands) as guiding information to reconstruct finer details in the lower resolution bands [8], [9]. This reduces the ambiguity associated with SR by anchoring the reconstruction in real, higher resolution measurements.

While diffusion models have gained traction in natural image SR [10], [11], [12], their application to remote sensing SR remains limited. In this work, we propose DiffFuSR, a two-step pipeline that begins by super-resolving Sentinel-2's 10-m RGB bands to 2.5 m using a diffusion model trained on harmonized aerial and Sentinel-2 data. Subsequently, a learned fusion network upscales the remaining spectral bands, using the super-resolved RGB as a spatial prior to ensure spectral integrity and mitigate hallucination.

We leverage large and diverse training datasets and carefully address domain-specific challenges. We utilize a diverse corpus from aerial NAIP imagery [3] and satellite-based WorldStrat data [13], despite their substantial spectral and radiometric differences from Sentinel-2. To bridge these gaps, we apply spectral harmonization techniques [3], which prove critical for generalization. Additionally, we incorporate blind kernel modeling [14] to simulate unknown degradations typical in real-world Sentinel-2 data. Evaluations using the OpenSR benchmark [15] and qualitative results show that our models yield significant performance improvements over current state-of-the-art baselines [2], [3], [7], [16].

To enhance the resolution of the remaining multispectral bands, we design a neural network (NN)-based fusion module

Received 7 November 2025; accepted 26 November 2025. Date of publication 1 December 2025; date of current version 11 December 2025. This work was supported by European Space Agency, Programme de Développement d'Expériences scientifiques (PRODEX) Program under Project 4000141281. *(Corresponding author: Muhammad Sarmad.)*

Muhammad Sarmad and Arnt-Børre Salberg are with the Norwegian Computing Center, 0314 Oslo, Norway (e-mail: sarmad92ee@gmail.com; salberg@nr.no).

Michael Kampffmeyer is with the Department of Physics and Technology, UiT The Arctic University of Norway, 9019 Tromsø, Norway, and also with the Norwegian Computing Center, 0314 Oslo, Norway.

Digital Object Identifier 10.1109/TGRS.2025.3638896

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:12:58 UTC from IEEE Xplore. Restrictions apply.

and train three instances of it for 10-, 20-, and 60-m bands. Each fusion model leverages the super-resolved 2.5-m RGB imagery as a high-frequency spatial prior, adopting a structure inspired by pansharpening principles. To enable supervised learning without external HR references, we simulate LR training pairs using the Wald protocol [17] on native Sentinel-2 data. Thisdesign ensures that the fusion task fully exploits the available spatial information. Evaluations show that our learned fusion models significantly outperform classical and learned baselinemethods in terms of quantitative metrics such as ERGAS, PSNR, and SSIM, as well as improved visual fidelity and spectral consistency.

In summary, DiffFuSR isa unified framework for super-resolving all Sentinel-2 bands to a common 2.5-m resolution. The key contributions are summarized as follows.

1) A novel diffusion-based SR framework for Sentinel-2 RGB bands, leveraging large-scale aerial and satellite datasets, enhanced with data harmonization and blind kernel modeling.
2) A new data fusion framework for multispectral SR that enhances the resolution of all Sentinel-2 bands (10, 20, and 60 m) to a unified 2.5-m spatial resolution using only native Sentinel-2 data, trained using Wald protocol simulations.
3) A comprehensive analysis of cross-domain SR challenges, demonstrating the critical importance of harmonization and degradation modeling through rigorous evaluation on the recently proposed OpenSR benchmark.

II. Background and Related Works

Early SR methods evolved from multi-image registration [18] and sparse representations [19] to pioneering deep learning approaches like SRCNN [20], which established CNNs as the standard for the task.

Pansharpening, i.e., methods that utilize the panchromatic band available in many satellites (e.g., Landsat and WorldView etc.) to enhance the resolution of simultaneously acquired multispectral data, has been a research topic for decades [21].

However, not all satellite sensors have a panchromatic band; hence, single-image SR has also become a central focus in remote sensing, particularly for enhancing the spatial resolution of Sentinel-2 imagery. Early work, such as Lanaras et al. [8], focused on enhancing the resolution of the RGB bands using CNNs trained with synthetic data. Later methods expanded to multispectral SR through data fusion approaches, combining multiple Sentinel-2 revisits [4] or external HR sources such as PlanetScope [6].

The generative power of generative adversarial networks (GANs) has also been explored for SR. Early works include the SRGAN approach by Ledig et al. [22], which was further improvedby Wang et al. [23]. The benefit of using generative methods like GANs for SR is the potential of creating photorealistic images, contrary to nongenerative methods, which tend to create too smooth results [22]. Recent advances in generative modeling have motivated the exploration of diffusion models for SR tasks [10], [11]. Wu et al. [14] applied conditional diffusion models for Sentinel-2 SR, enhancing only the RGB bands by leveraging paired SPOT 6/7 imagery. However, their approach remains constrained to RGB bands, and the reliance on paired HR datasets limits its direct scalability to the remaining Sentinel-2 bands.

Efforts such as the WorldStrat dataset [13] and the OpenSR benchmark [15] have provided valuable resources for training and evaluating remote sensing SR models under real-world conditions. Wolters et al. [7] leveraged NAIP aerial imagery for generative SR but faced challenges due to spectral and radiometric inconsistencies between NAIP and Sentinel-2, leading to domain gaps that degraded the performance. The SEN2NAIP dataset and accompanying work by Aybar et al. [3] introduced a harmonized degradation model to better align NAIP images with Sentinel-2 characteristics, proving crucial for cross-sensor learning. Donike et al. [2] employed latent diffusion models trained on SEN2NAIP, demonstrating that operating in the latent space can significantly reduce resource consumption. They focus solely on super-resolving the 10-m Sentinel-2 bands.

After performing SR on the 10-m bands, pansharpening techniques can be applied to fuse the super-resolved image with the LR multispectral bands (e.g., 20 and 60 m) [24]. Classical methods such as GS pansharpening [25] can be effective but may prove insufficient or introduce spectral distortions when attempting to fuse super-resolved RGB images (acting as a pseudo-panchromatic band) with Sentinel-2's remaining multispectral bands, especially when targeting a significant resolution enhancement to 2.5 m [24]. Lanaras et al. [8] demonstrated that NNs can perform such fusion more effectively than classical methods, mitigating some distortions. However, their method was limited to enhancing bands to a 10-m ground sampling distance (GSD) instead of a higher resolution of 2.5-m GSD.

The modular framework presented in this article significantly extends previous work by jointly addressing SR of all Sentinel-2 bands to a unified 2.5 m resolution. First, we super-resolve the RGB bands using an efficient diffusion model trained on harmonized data, ensuring robust generalization. Second, a learned fusion network, inspired by pansharpening principles but trained for superior spectral fidelity, upscales the remaining 10-, 20-, and 60-m Sentinel-2 bands. While prior works address isolated components of this problem, to the best of our knowledge, this is the first unified approach that performs blind SR across all Sentinel-2 bands and resolutions.

III. Method

First, in Section III-A, we describe a conditional denoising diffusion model to super-resolve the RGB bands to 2.5-m GSD, trained on harmonized synthetic data to ensure robust generalization. Second, in Section III-B, we introduce a learned multispectral fusion network that leverages the super-resolved RGB output to upscale the remaining Sentinel-2 bands to the target GSD of 2.5 m. The overall pipeline is illustrated in Fig. 1.

A. SR of RGB Bands Using Conditional Denoising Diffusion Models

We formulate the RGB SR task as conditional generation using a denoising diffusion probabilistic model (DDPM)

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:12:58 UTC from IEEE Xplore. Restrictions apply.

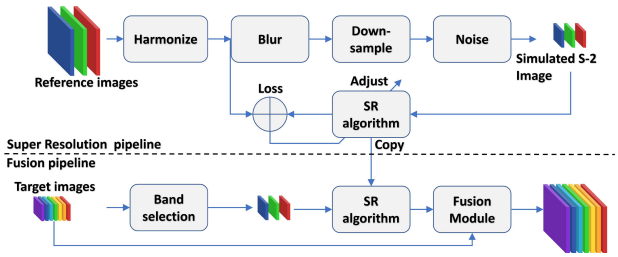


Fig. 1. Overview of the proposed pipeline combining RGB SR and multispectral fusion. The top branch illustrates the training of a denoising diffusion model using synthetic LR RGB images generated through harmonization, blurring, downsampling, and noise addition. The model learns to reconstruct HR RGB images. The bottom branch depicts the fusion process: multispectral Sentinel-2 bands are fused with the super-resolved RGB output via a learned fusion module to produce full-resolution 12-band output.

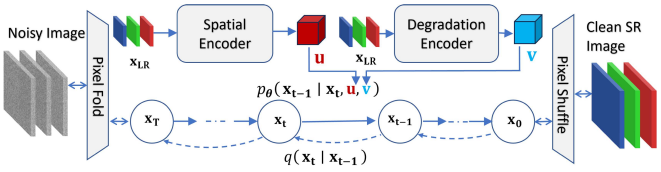


Fig. 2. Architecture of the proposed conditional denoising diffusion model for RGB SR. The model learns to reconstruct an HR RGB image $\mathbf{x}_0$ from a noisy latent $\mathbf{x}_T$ via a denoising process modeled as a Markov chain. The input image is first degraded with Gaussian noise and spatially rearranged using a pixel-folding operation. A spatial encoder extracts contextual features $\mathbf{u}$ from the LR input, while a degradation encoder produces a global degradation embedding $\mathbf{v}$. Both $\mathbf{u}$ and $\mathbf{v}$ are used to condition the denoising network at each reverse time step t. The sequence $\mathbf{x}_T \to \cdots \to \mathbf{x}_0$ is denoised through a learned reverse diffusion process. The final output is reshaped using a pixel shuffle operation to recover the super-resolved RGB image.

inspired from [14]. The goal is to generate an HR RGB image $\mathbf{x}_0 \in \mathbb{R}^{C\times H\times W}$, conditioned on an LR observation $\mathbf{x}_{\text{LR}} \in \mathbb{R}^{C\times H'\times W'}$, where $H' < H$ and $W' < W$. The model learns to reverse a Gaussian noise corruption process added to $\mathbf{x}_0$, using the fixed $\mathbf{x}_{\text{LR}}$ to create multiple conditioning inputs. The SR pipeline consists of a forward diffusion process and a reverse denoising process (Fig. 2).

1) Forward Diffusion Process: Following [26], the forward process is a Markov chain that progressively adds Gaussian noise to a "clean" input image $\mathbf{x}_0$ over T steps:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\mathbf{x}_t;\ \sqrt{\alpha_t}\mathbf{x}_{t-1}, (1-\alpha_t)\,\mathbf{I}\right) \tag{1}$$

with a predefined noise schedule $\{\alpha_t\}_{t=1}^{T}$. This defines the marginal distribution

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_t;\ \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\,\mathbf{I}\right) \tag{2}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t}\alpha_s$. At $t = T$. The sample $\mathbf{x}_T$ approximates a standard Gaussian.

2) Reverse Denoising Process: The reverse denoising process is modeled as a Markov chain conditioned on the LR input $\mathbf{x}_{\text{LR}}$. Starting from Gaussian noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, this process iteratively denoises a sequence of latent variables $\mathbf{x}_T, \ldots, \mathbf{x}_0$ according to

$$p_{\boldsymbol{\theta}}(\mathbf{x}_{0:T}|\mathbf{u}, \mathbf{v}) = p(\mathbf{x}_T)\prod_{t=1}^{T} p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{u}, \mathbf{v}) \tag{3}$$

where each reverse transition $p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{u}, \mathbf{v})$ is modeled as a Gaussian distribution

$$p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{u}, \mathbf{v}) = \mathcal{N}\left(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \mathbf{u}, \mathbf{v}), \sigma_t^2\mathbf{I}\right). \tag{4}$$

The variance σ_t^2 is typically predefined, while the mean $\boldsymbol{\mu}_{\boldsymbol{\theta}}$ is predicted by a denoising network, commonly a U-Net architecture [27], conditioned on the noisy latent $\mathbf{x}_t$, the timestep t, and the fixed context $\mathbf{u}, \mathbf{v}$. The integration of $\mathbf{u}$ and $\mathbf{v}$ allows the model to incorporate both spatial structural details and degradation-specific behavior, thereby enabling accurate and blind SR.

The conditioning inputs $\mathbf{u}$ and $\mathbf{v}$ are fixed throughout the reverse process and derived from the observed LR image $\mathbf{x}_{\text{LR}}$ as follows.

1) The spatial feature map $\mathbf{u} = f_{\boldsymbol{\phi}}(\mathbf{x}_{\text{LR}})$ is extracted by an encoder $f_{\boldsymbol{\phi}}$. It captures coarse structural information, such as edges and object geometries from $\mathbf{x}_{\text{LR}}$. The spatial feature map, $\mathbf{u}$, is concatenated at the input of the UNet.
2) The global degradation embedding $\mathbf{v} = g_{\boldsymbol{\psi}}(\mathbf{x}_{\text{LR}})$ summarizes the underlying blur, noise, and downsampling characteristics of $\mathbf{x}_{\text{LR}}$. This embedding is computed using a separate encoder $g_{\boldsymbol{\psi}}$, trained via contrastive learning [28].

3) Network Architecture: The proposed diffusion model is comprised of three main components (Fig. 2): 1) a *denoising U-Net* that predicts the clean HR image from noisy input; 2) a *spatial encoder* that extracts contextual features from the LR image; and 3) a *degradation encoder* trained with contrastive learning to summarize the degradation characteristics. We follow the design by Wu et al. [14]:

a) Denoising U-Net: The core network $h_{\boldsymbol{\theta}}$ is a conditional U-Net that predicts the clean image $\mathbf{x}_0$ given a noisy input $\mathbf{x}_t$, timestep t, and conditioning vectors $\mathbf{u}, \mathbf{v}$. To improve computational efficiency without losing information, we apply a *pixel-folding* operation to $\mathbf{x}_t$ at the input [14]. This reduces spatial resolution by a factor r while increasing the number of channels by r^2. Formally, this maps an image of dimension $C \times H \times W$ to $(C \cdot r^2) \times (H/r) \times (W/r)$. The folded input is passed through the denoisingU-Net [14].After denoising, the output is unfolded back to the original spatial resolution using the pixel shuffle operation [29].

To inform the network of the current step in the reverse denoising process, a timestep embedding is injected at each residual block. The scalar timestep $t \in \{1, \ldots, T\}$ is first mapped to a sinusoidal positional encoding

$$\boldsymbol{\kappa}(t) = [\sin(\omega_1 t), \cos(\omega_1 t), \ldots, \sin(\omega_d t), \cos(\omega_d t)]^T$$

and then passed through a multilayer perceptron (MLP) to produce the embedding $\mathbf{t}_e$. This embedding is added to intermediate feature maps throughout the U-Net [14].

b) Spatial Encoder: The LR input $\mathbf{x}_{\text{LR}}$ is processed by a spatial encoder $f_{\boldsymbol{\phi}}$, implemented as an RRDB-Net consisting of 23 residual-in-residual dense blocks (RRDBs) with 64 feature channels [23]. The encoder outputs a spatial feature map $\mathbf{u} = f_{\boldsymbol{\phi}}(\mathbf{x}_{\text{LR}})$, which serves as a global spatial conditioning signal for the denoising model.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:12:58 UTC from IEEE Xplore. Restrictions apply.

Rather than being fused at multiple layers, the spatial feature map $\mathbf{u}$ is concatenated with the input of the U-Net denoiser at the first layer only. This provides the network with high-level spatial context from the LR image while keeping the fusion interface lightweight. During training, the decoder (a lightweight CNN, not shown in Fig. 2) is attached to the encoder to predict a coarse HR reconstruction $\widehat{\mathbf{x}}_0$, supervised with an ℓ_1 consistency loss. The decoder is discarded during inference.

c) Degradation Encoder: To model the implicit degradation of the LR image, we use a lightweight CNN $g_{\boldsymbol{\psi}}$ that maps an LR image patch $\mathbf{x}_{\mathrm{LR}}$ to a global degradation vector $\mathbf{v} = g_{\boldsymbol{\psi}}(\mathbf{x}_{\mathrm{LR}})$. The network consists of several convolutional layers, followed by global average pooling and a projection head. It is trained with a contrastive objective [30], wherethe goal is to train a model such that given two crops $\mathbf{x}_q$, $\mathbf{x}_k$ from the same LR image (i.e., with the same degradation), their embeddings $\mathbf{v}_q$, $\mathbf{v}_k$ should be similar, while embeddings from different images (assumed to have different degradations) should be dissimilar. Under this setting, the model learns about the degradation in an unsupervised manner.

The learned degradation vector $\mathbf{v}$ is injected into each residual block in the U-Net using a DAConv layer [14]. In this approach, the convolutional filters are dynamically modulated based on the conditioning vector, allowing the network to adapt to different degradation types during inference.

4) Training Objective: The model is trained end-to-end using a combination of three loss terms.

1) *Denoising Loss:* The diffusion model is trained to predict the clean HR image $\mathbf{x}_0$ from its noisy version $\mathbf{x}_t$, conditioned on spatial and degradation features using the ELBO loss [26]

$$\mathcal{L}_{\mathrm{elbo}} = \|\mathbf{x}_0 - h_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \mathbf{u}, \mathbf{v})\|_1. \tag{5}$$

2) *Consistency Loss:* An auxiliary decoder, $\xi(\cdot)$, is attached to the spatial encoder $f_{\boldsymbol{\phi}}$, which processes the LR input $\mathbf{x}_{\mathrm{LR}}$. The decoder predicts a coarse HR estimate $\widehat{\mathbf{x}}_0 = \xi(f_{\boldsymbol{\phi}}(\mathbf{x}_{\mathrm{LR}}))$, and the output is compared against the ground truth $\mathbf{x}_0$ using an L1 loss

$$\mathcal{L}_{\mathrm{consis}} = \|\widehat{\mathbf{x}}_0 - \mathbf{x}_0\|_1. \tag{6}$$

This encourages the spatial encoder to learn features that are aligned with the HR target. The decoder is only used during training and discarded at inference.

3) *Contrastive loss:* We use the InfoNCE loss for training [14]. Given two patches $\mathbf{x}_{\mathrm{LR}}$ and $\mathbf{x}^+_{\mathrm{LR}}$ from the same image (positive pair), and a set of negatives $\{\mathbf{x}^-_{\mathrm{LR}}\}$ from different images in the batch, the contrastive loss is defined as

$$\mathcal{L}_{\mathrm{contrast}} = -\log \frac{\exp\left(\mathrm{sim}\left(\mathbf{v}, \mathbf{v}^+\right)/\tau\right)}{\sum_j \exp\left(\mathrm{sim}\left(\mathbf{v}, \mathbf{v}_j\right)/\tau\right)} \tag{7}$$

where $\mathbf{v} = g_{\boldsymbol{\psi}}(\mathbf{x}_{\mathrm{LR}})$, $\mathbf{v}^+ = g_{\boldsymbol{\psi}}(\mathbf{x}^+_{\mathrm{LR}})$, $\mathbf{v}_j$ includes all negative samples, τ is a temperature hyperparameter, and $\mathrm{sim}(\cdot,\cdot)$ denotes cosine similarity, i.e., $\mathrm{sim}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a}^T\mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|}$.

The total training loss is then a weighted sum

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{elbo}} + \mathcal{L}_{\mathrm{consis}} + \lambda_{\mathrm{contrast}} \cdot \mathcal{L}_{\mathrm{contrast}} \tag{8}$$

where the constant $\lambda_{\mathrm{contrast}}$ is a weighting factor that determines the contribution of the contrastive loss $\mathcal{L}_{\mathrm{contrast}}$ relative to the ELBO loss $\mathcal{L}_{\mathrm{elbo}}$ and the consistency loss $\mathcal{L}_{\mathrm{consis}}$. A small default value, such as $\lambda_{\mathrm{contrast}} = 0.01$, ensures that the contrastive objective improves representation learning without overwhelming the main generative and consistency objectives.

5) Sampling: At inference time, an HR image is sampled by starting from $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$ and iteratively applying the learned denoising model conditioned on $\mathbf{u}$ and $\mathbf{v}$. The final output $\mathbf{x}_0$ is returned as the super-resolved RGB image.

6) Simulating Sentinel-2 Images: We adopt a simple, statistics-based harmonization model using per-channel power-law transformations to generate synthetic Sentinel-2-like images from NAIP images. This is achieved by a two-stage transformation: 1) spectral harmonization to match Sentinel-2 reflectance distributions and 2) randomized degradation to simulate optical and sensor effects such as blur and noise. This strategy enables fully self-supervised training and supports blind SR without reliance on paired ground truth.

a) Harmonization Model: Harmonization adjusts the spectral distribution of NAIP imagery to resemble the Sentinel-2 RGB bands, enabling more effective cross-domain learning. We perform per-channel gamma correction using a power-law transformation

$$\widehat{\mathbf{x}}^c_{NP} = \left(\frac{\mathbf{x}^c_{NP}}{k}\right)^{\frac{1}{\gamma_c}} \cdot k \tag{9}$$

where $\mathbf{x}^c_{NP}$ denotes band c of the NAIP image, $k = 255$ denotes the dynamic range, and γ_c is the channel-specific gamma value. These gamma values are derived empirically by aligning NAIP reflectance distributions to co-registered Sentinel-2 imagery. Optimal gamma values are computed for each channel across a large corpus of image pairs [3]. A multivariate normal model is then fit to the resulting gamma vectors, and the median values are used to define a global correction profile. This ensures consistent spectral alignment during inference without the need for image-specific tuning.

b) Degradation Model: To simulate Sentinel-2-like degradations, we apply a blind degradation strategy that combines randomized blur, downsampling, and additive noise. The degraded image is modeled as

$$\widehat{\mathbf{x}}_{S2} = \left[\Phi_{\boldsymbol{\rho}}\left(\widehat{\mathbf{x}}_{NP}\right)\right]_{\downarrow} + \mathbf{n} \tag{10}$$

where $\Phi_{\boldsymbol{\rho}}(\cdot)$ is a Gaussian blur operator parameterized by $\boldsymbol{\rho}$, $\downarrow$ denotes bicubic downsampling, and $\mathbf{n}$ is additive Gaussian noise.

c) Blur Model: To emulate optical blur, we use both isotropic and anisotropic Gaussian kernels, with parameters sampled randomly for each training image. We select a fixed kernel size of 21×21. During training, we use an anisotropic Gaussian blur type, whereas for validation, we use an isotropic Gaussian. We sample the smoothing parameter σ from the range $\sigma \in [2.0, 4.0]$, the eigenvalue range of the anisotropic Gaussian kernel as $\lambda_1, \lambda_2 \in [0.2, 4.0]$, and the rotation angle as $\theta \in [0, \pi]$.

d) Noise Model: Sensor noise is modeled as additive Gaussian noise, i.e.,

$$\mathbf{n} \sim \mathcal{N}\left(0, \sigma_n^2 \mathbf{I}\right), \quad \sigma_n \in [0.0, 25.0] \tag{11}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:12:58 UTC from IEEE Xplore. Restrictions apply.

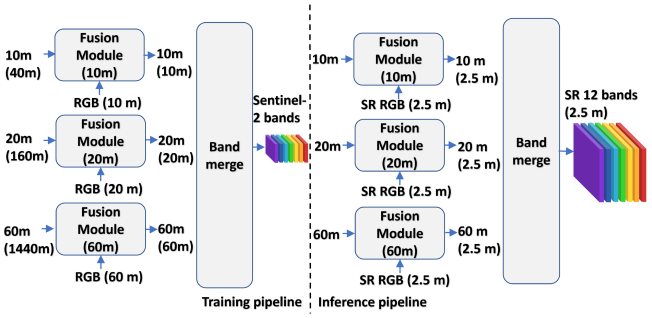


Fig. 3. Architecture of the multispectral fusion pipeline. At test time, each resolution group of 10-, 20-, and 60-m bands is processed by a dedicated fusion module that receives the native-resolution multispectral inputs along with the RGB image super-resolved to 2.5 m. These modules independently reconstruct each group at 2.5-m GSD. The outputs are then merged to produce a consistent 12-band super-resolved Sentinel-2 image. This process is applied during both training (using synthetic degraded inputs) and inference (using native-resolution bands), with shared architecture and supervision strategy across all groups. The scaling factor for degraded input for training is computed using the Wald protocol. For example, for the 60-m group, we aim to learn the mapping from 60 to 2.5 m given a 2.5-m SR guidance signal. This corresponds to a 24× scale factor. During training, 60-m bands are degraded by this factor, producing input resolutions of 1440 m. At training time, the network reconstructs the degraded signal to 60-m GSD, supervised by the corresponding 60-m ground-truth reference.

with σ_n sampled randomly for each image during training.

e) Blind Conditioning: The same degraded image is passed to both the spatial encoder f_{ϕ} and the degradation encoder g_{ψ}. The randomized, unsupervised nature of the degradation process enables the model to generalize to unknown test-time corruptions. In particular, the degradation representation $\mathbf{v}$ is learned entirely without supervision of degradation parameters, using contrastive learning to capture differences in blur, noise, and downsampling implicitly.

B. Multispectral Band Fusion

To produce a consistent 2.5-m resolution across all Sentinel-2 bands, we design a learnable fusion approach that reconstructs HR multispectral bands by injecting spatial details from the super-resolved RGB image, $\mathbf{x}_0$ (Fig. 3). This is formulated as a guided upsampling problem, where the RGB image provides high-frequency structural cues and the LR multispectral bands contribute the spectral content.

Let $\mathbf{X}_{\mathrm{LR}}$ denote multispectral data at its native resolution (10, 20, and 60 m). The objective of the fusion module, $F_{\nu}(\cdot,\cdot)$, is to learn a mapping

$$\widehat{\mathbf{X}}_{\mathrm{HR}} = F_{\nu}\left(\mathbf{X}_{\mathrm{LR}}, \mathbf{x}_0\right)$$

where $\widehat{\mathbf{X}}_{\mathrm{HR}}$ is the reconstructed data at 2.5-m GSD, andν denotes the parameters of the fusion model. In practice, the fusion pipeline contains three different modules, one for each band group (10, 20, and 60 m), which are trained and tested simultaneously but completely independently of each other (Fig. 3).

The three proposed fusion pipelines are based on the generalized Laplacian pyramid (GLP) with modulation transfer function (MTF) filters and a regression injection model

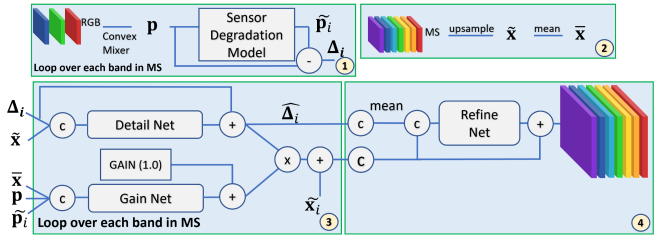


Fig. 4. Architecture of the proposed GLP-inspired [31] fusion module. Each model variant for the 10-, 20-, and 60-m branches contains approximately 202k trainable parameters. The symbols denote basic operations: "+" indicates element-wise addition, "−" indicates subtraction, "×" indicates element-wise multiplication, and "c" indicates channel-wise concatenation.

(MTF-GLP) [31], [32], as illustrated in Fig. 4. First, the panchromatic image is low-pass filtered by band-specific MTF filters and then downsampled and upsampled. Subtracting this result from the original panchromatic image yields the spatial details for injection. The injection coefficients, which control how much detail is added to each multispectral band, are estimated by a regression procedure.We embed the main components of MTF-GLP into the NN architecture. The per-band low-pass filtering is implemented as fixed convolutional filters mimicking the sensor's MTF, corresponding to the panchromatic degradation step in the classical model. GLP-style detail extraction is realized by subtracting the low-pass filtered panchromatic image from the original panchromatic image, with a residual detail network learning only the correction beyond the classical detail term. The regression-based injection mechanism is a gain estimator, initialized around unity, which adapts the detail injection for each band. Finally, a shallow refinement network enforces interband consistency, compensating for nonlinear effects not captured by linear regression.

1) Training Strategy and Degradation Simulation: The fusion model is trained in a fully self-supervised manner using only native Sentinel-2 data. we adopt a Wald protocoltraining strategy where the input bands are downsampled from their native resolution, and the original band serves as the supervision target.

To simulate realistic downsampled imagesduring training, each input band undergoes a degradation process

$$\mathbf{X}_{\mathrm{LR}}^{\downarrow} = \left[\Phi_{\sigma}\left(\mathbf{X}_{\mathrm{LR}}\right)\right]_{\downarrow} \tag{12}$$

where $\Phi_{\sigma}(\cdot)$ is an optional isotropic Gaussian blur applied to emulate sensor response and $\downarrow$ denotes the boxcar filter, implemented as average pooling with stride corresponding to the target scale.

2) Network Implementation: The fusion model $F_{\nu}(\cdot,\cdot)$ is instantiated via a separate network for each resolution group (10, 20, and 60 m), with approximately 200k parameters per model.

The model is trained to minimize an ℓ_1 reconstruction loss

$$\mathcal{L}_{\mathrm{fusion}} = \left\| \widehat{\mathbf{X}}_{\mathrm{LR}} - \mathbf{X}_{\mathrm{LR}} \right\|_1$$

where

$$\widehat{\mathbf{X}}_{\mathrm{LR}} = \mathcal{F}_{\nu}\left(\mathbf{X}_{\mathrm{LR}}^{\downarrow}, \mathbf{x}_{\mathrm{LR}}\right) \tag{13}$$

is the predicted fused output.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:12:58 UTC from IEEE Xplore. Restrictions apply.

a) Detailed Architecture: The details of architecure given in Fig. 4) are as follows.

1) Convex Mixer: The bands of the Sentinel-2 RGB, $\mathbf{x}_{LR}$, are combined into a panchromatic proxy

$$\mathbf{p} = \sum_{c=1}^{3} w_c \, \mathbf{x}_{\mathrm{LR}}, \qquad w_c \geq 0, \ \sum_{c} w_c = 1$$

where $\{w_c\}$ is the softmax-constrained, learnable convex weight.

2) Sensor Degradation Model: For each band i, a Gaussian MTF blur Φ_{σ_i} with reflection padding, followed by area downsampling and bicubic upsampling $(\mathcal{D}_{r_i}, \mathcal{U}_{r_i})$, produces the low-pass panchromatic image approximation

$$\widetilde{\mathbf{p}}_i = \mathcal{U}_{r_i}\left(\mathcal{D}_{r_i}\left(\left(\Phi_{\sigma_i}(\mathbf{p})\right)\right)\right)$$

and the high-frequency details that need to be injected in the multispectral bands for each band are given as

$$\mathbf{\Delta}_i = \mathbf{p} - \widetilde{\mathbf{p}}_i.$$

Here, the scaling factor r_i denotes the ratio between the native GSD of the ith multispectral band and the panchromatic proxy resolution. In practice, r_i determines the downsampling and upsampling ratio applied in the simulated sensor degradation model (Fig. 4). Specifically

$$r_i = \frac{\text{native GSD of band } i}{\text{target PAN GSD}}.$$

For the Sentinel-2 configuration, we use three dedicated fusion branches corresponding to the 10-, 20-, and 60-m resolution groups. We use $r_i = 4, 8,$ and 24 for the 10-, 20-, and 60-m bands, respectively. Each branch reconstructs its respective multispectral group to the 2.5-m target resolution, guided by the super-resolved RGB proxy. Each blur kernel Φ_{σ_i} is initialized from the nominal MTF value of the corresponding Sentinel-2 band, converted to a Gaussian standard deviation σ_i as a function of the Nyquist frequency and the scale ratio r_i.

3) DetailNet: Each band's detail is refined by a small residual network that operates on the concatenation of the GLP detail and the upsampled multispectral band group that is input to the fusion module, $\widetilde{\mathbf{x}}$ (Fig. 4)

$$\widehat{\mathbf{\Delta}}_i = \mathbf{\Delta}_i + \mathrm{DetailNet}\left([\mathbf{\Delta}_i, \ \widetilde{\mathbf{x}}]\right)$$

where DetailNet consists of a 3×3 convolution, LeakyReLU, two RRDB blocks, and a linear 3×3 conv that outputs a one-channel residual map. The final layer has no activation to retain the signed nature of the learned residual.

4) GainNet: A per-band residual gain map is estimated from the concatenation of the panchromatic proxy, its band-specific low-pass approximation, and the mean of the upsampled multispectral inputs

$$g_i = 1 + \mathrm{GainNet}\left(\left[\ \mathbf{p}, \ \widetilde{\mathbf{p}}_i, \ \bar{\mathbf{x}}\ \right]\right)$$

where $\bar{\mathbf{x}}$ denotes the per-pixel mean of the upsampled multispectral group $\widetilde{\mathbf{x}}$. GainNet consists of a shared two-layer convolutional subnetwork: a 3×3 convolution with LeakyReLU activation (0.2 slope) followed by a 1×1 convolution that outputs a single-channel residual map Δg_i. The final gain is formulated as $g_i = 1 + \Delta g_i$, providing an unbounded residual gain around unity that scales the injected detail adaptively for each band.

5) RefineNet and Fusion: The bandwise fusion applies the gain-modulated detail injection

$$\mathbf{y}_i = \widetilde{\mathbf{x}}_i + g_i \cdot \widehat{\mathbf{\Delta}}_i$$

where $\widetilde{\mathbf{x}}_i$ is the ith input multispectral band. The output $\{\mathbf{y}_i\}$ is then stacked into into $\mathbf{Y}$. A shallow refinement network then enforces interband consistency using the concatenation $[\mathbf{Y}, \ \bar{\mathbf{\Delta}}_i]$, where $\bar{\mathbf{\Delta}}_i$ is the mean refined across all detail bands $\{\mathbf{\Delta}_i\}$

$$\widehat{\mathbf{X}}_{\mathrm{HR}} = \mathbf{Y} + \mathrm{RefineNet}\left(\left[\mathbf{Y}, \ \bar{\mathbf{\Delta}}_i\right]\right).$$

RefineNet consists of a 3×3 convolution with LeakyReLU activation, two residual blocks, and a final linear 3×3 layer that outputs a residual correction added back to $\mathbf{Y}$.

b) Inference: In inference, the fusion model receives inputs at their original resolutions (10, 20, or 60 m), but the native RGB input is replaced by a 2.5-m GSD super-resolved RGB image. This new input is low-pass filtered with the same per-band MTF kernels used in training to ensure frequency consistency. The network must generalize zero-shot to this enhanced spatial prior, as the super-resolved image is not used during the training phase [8].

IV. Experiments and Results

A. Datasets

To train and evaluate the proposed SR method, we use the following datasets.

1) SEN2NAIP Dataset: The SEN2NAIP dataset [3] consists of two components.

1) *Cross-Sensor Dataset*: Paired imagery from Sentinel-2 and National Agriculture Imagery Program (NAIP), suitable for supervised learning tasks involving cross-sensor translation.
2) *Synthetic Dataset*: Standalone NAIP images accompanied by transformation parameters for simulating Sentinel-2-like imagery. These parameters are based on four different histogram-based harmonization strategies.

2) WorldStrat Dataset: The WorldStrat dataset [13] contains HR RGB imagery acquired from the SPOT satellite. The data are globally distributed and tile-based, covering diverse geographic regions and land cover types.

3) Sentinel-2 Data: The Sentinel-2 data used in this work consists of Level-2A products from the Sentinel-2 mission under the Copernicus program. These products provide 12 spectral bands at spatial resolutions of 10, 20, and 60 m. All imagery has been curated to be cloud-free and atmospherically corrected, with spatial and temporal alignment across samples.

B. SR Model Training

We train models using three primary data sources: airborne (NAIP), satellite-based (WorldStrat), and multispectral (Sentinel-2). These datasets are integrated into a unified diffusion-based SR pipeline.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:12:58 UTC from IEEE Xplore. Restrictions apply.

1) Model Architecture: The model architecture follows the design of Wu et al. [14] and comprises the following.

1) A degradation encoder for contrastive representation learning. (2.6 M parameters)
2) A spatial encoder–decoder that is implemented using an RRDB-Net encoder with 23 residual blocks and 64 feature channels.(16.7 M parameters)
3) A diffusion UNet with 5 residual blocks and channel multipliers {1, 2, 2, 4, 8}. (148.3 M parameters)
4) Diffusion configuration: we use 1000 diffusion steps, with a cosine noise schedule [14]. The model predicts the original clean image $\mathbf{x}_0$ directly.
5) Exponential moving average of weights with a decay rate of 0.999.

2) Training Configuration: All models were trained for 1000 epochs using the Adam optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$) with an initial learning rate of 1×10^{-4}, decayed by a factor of 0.5 at epochs 200, 400, 600, and 800. We used a batch size of 8 and an image patch size of 256×256. Data augmentation included random cropping and flipping.

3) NAIP (Harmonized and Unharmonized): Training used samples from the synthetic set of SEN2NAIP [3]. In the harmonized variant, on-the-fly histogram harmonization was applied during data loading. For unharmonized training, this step was omitted. Both configurations used four-channel multispectral inputs. Degradation modeling applied anisotropic Gaussian blur (kernel size 21 and $\sigma \in [2.0, 4.0]$) and additive Gaussian noise with standard deviation 25.

4) WorldStrat: A separate model was trained on globally distributed tiles from the WorldStrat dataset [13]. As with NAIP, four-channel input was used and the same model architecture and optimizer configuration were applied.

C. Fusion Model Training

1) Model Architecture: The proposed fusion model is implemented as three dedicated MTFNetFusion modules, one per resolution group (10, 20, and 60 m).

1) A *10-m branch* that fuses RGB and near infrared bands (bands 2–4 and 8). (202k parameters)
2) A *20-m branch* that fuses the red edge and short wave infrared bands (bands 5, 6, 7, 8A, 11, and 12). (202k parameters)
3) A *60-m branch* that fuses coastal aerosol and water vapor bands (bands 9 and 10). (202k parameters)

Each branch follows an MTF-GLP design, as specified in Section III-B.

2) Training Details: Three models are trained using separate optimizers for each spatial branch (10, 20, and 60 m), each using Adam with a learning rate of 1×10^{-4}.

3) Validation Metric: Validation is performed using ERGAS at each resolution.Additionally, multiple pansharpening baselines are computed for comparison.

4) Inference Time: The inference time for the diffusion model is 1.74 s, and for the fusion model, it is 0.0161 s.

D. Baseline Methods

Existing state-of-the-art methods do not directly address the task of super-resolving all Sentinel-2 bands to a unified resolution. Since DiffFuSR is modular, we evaluate its components in two stages. First, we compare its RGB SR module against established RGB-specific SR baselines. Second, we evaluate the fusion module separately by comparing it to a suitable multispectral fusion baseline that combines super-resolved RGB with the remaining bands.

1) SR Baselines: Theproposed SR approach is evaluatedagainst a diverse set of existing models.The *OpenSR baseline* [3] serves as a strong reference point, particularly due to its use of degradation-based training on NAIP data. The *LDM_Baseline* [2] employs a latent diffusion model, applying generative priors to RGB upsampling. *Superimage* adopts a pixel-wise regression framework optimized for spectral fidelity. We compare against *Satlas* [7], a GAN-based model trained on cross-domain imagery.Additionally, we compare with *SR4RS* [16], a regression-based model targeting RGB SR via auxiliary feature learning.

2) Fusion Model Baseline: To benchmark the effectiveness of the proposed fusion network (GLP-NN), we implement GS pansharpening [9], the intensity-hue-saturation (IHS) transform, principal component analysis (PCA), and the classical GLP with MTF filters and full-scale regression injection model (GLP) method [31]. Furthermore, following [21], we include their state-of-the-art FusionNet as a learning-based baseline. Although FusionNet was originally designed for generic pansharpening, we retrain it under the same conditions as our proposed fusion module to ensure a fair comparison.

E. Metrics

1) Super Resolution: We evaluate the proposed SR model using the OpenSR-test benchmark [15], which offers cross-sensor datasets (NAIP, SPOT, VenμS, SPAIN CROPS, and SPAIN URBAN) with accurately aligned Sentinel-2 LR and HR reference imagery. The benchmark assesses performance through a set of intuitive metrics, grouped into three categories.

Consistency evaluates whether the super-resolved image, when downsampled, preserves key properties of the original Sentinel-2 input. It includes: 1) *reflectance*, measuring radiometric similarity via chosen distance metric; 2) *spectral*, quantifying preservation of spectral signatures through spectral angle distance (SAD); and 3) *spatial*, assessing alignment using phase correlation.

Synthesis captures how much meaningful high-frequency detail the model introduces beyond simple interpolation.

Correctness analyzes whether the generated details are realistic. It includes: 1) *hallucinations*, quantifying spurious details not present in the true HR reference; 2) *omissions*, measuring genuine HR details that the model failed to recover; and 3) *improvements*, reflecting correctly added fine-scale structures that were not visible in the LR input.

The distance metric chosen for calculation can be either normalized difference distance, LPIPS [33], or CLIP distance [34]. For full mathematical definitions and implementation details, we refer to the OpenSR-test paper [15].

2) Fusion Pipeline: To evaluate the quality of the SR fusion outputs in the absence of ground-truth HR Sentinel-2 imagery, we adopt a no-reference validation strategy. Specifically, we

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:12:58 UTC from IEEE Xplore. Restrictions apply.

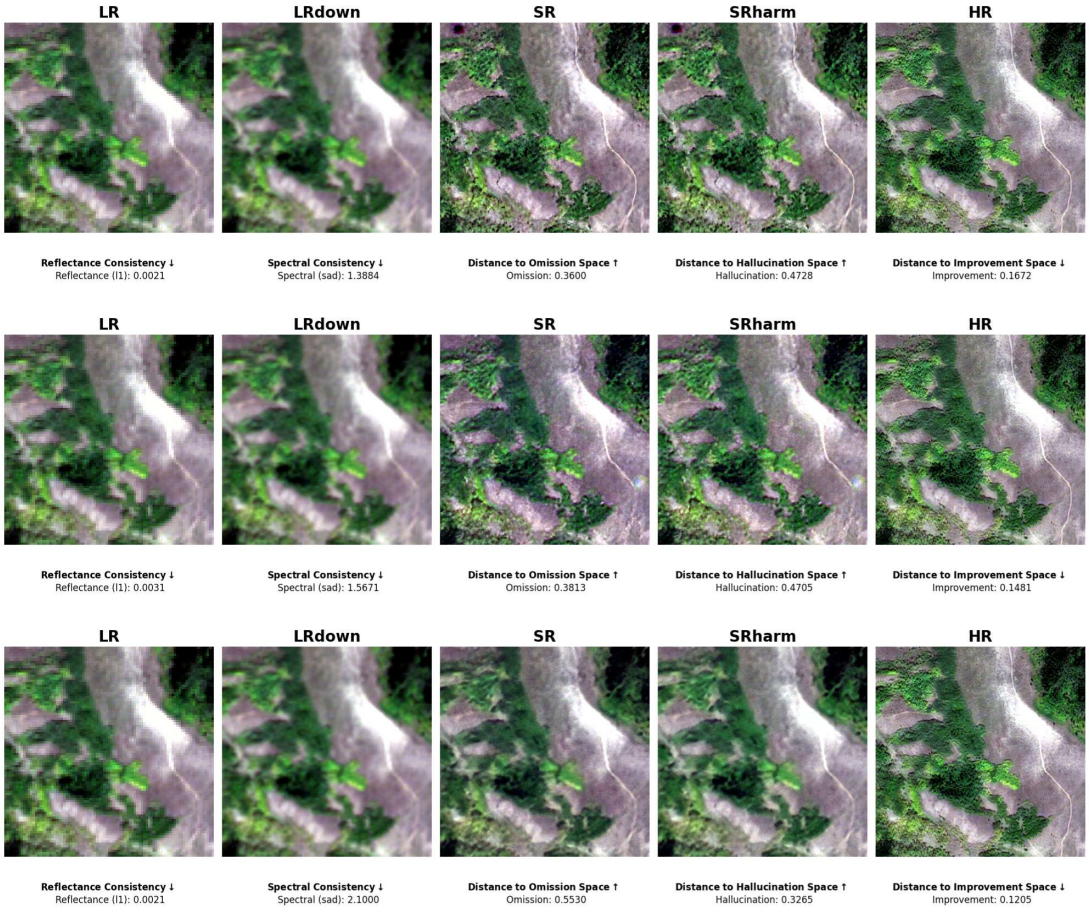


Fig. 5. Qualitative comparison of RGB SR results (vegetated scene). Rows correspond to models trained on harmonized NAIP, unharmonized NAIP, and WorldStrat. The top subrow shows image outputs: LR input, bicubic downsampled variant, SR result, harmonized SR, and ground truth. Bottom subrow shows reflectance consistency (L1), spectral consistency (SAD), and distances to omission, hallucination, and improvement spaces. The harmonized NAIP model yields the best visual and quantitative performance in this vegetated context.

downsample the SR outputs back to the original Sentinel-2 resolution and perform comparisons against the native Sentinel-2 inputs. Evaluations are carried out using the following metrics: coefficient of determination (R^2), cross correlation, structural similarity index (SSIM) [35], peak signal-to-noise ratio (PSNR) [35], mean squared error (MSE), and ERGAS (Relative Dimensionless Global Error in Synthesis) [17].

F. Qualitative Evaluation

1) Super Resolution: Figs. 5 and 6 illustrate qualitative comparisons of the proposed SR model trained on three different datasets: harmonized NAIP, unharmonized NAIP, and WorldStrat. Each figure is organized into three major rows (one per dataset), where eachrow shows images, including the LR input, downsampled version, SR output, harmonized SR output, and the HR reference.

In Fig. 5, which features a forested scene with green tree cover and bare land, the harmonized NAIP model clearly yields the sharpest and most visually realistic reconstruction. Both the SR image and the harmonized version closely resemble the HR reference, with fine details preserved. This is supported by the quantitative values, where reflectance and spectral consistency metrics are lowest (L1: 0.0021 and SAD: 1.3884) and omission and hallucination errors are moderate. The unharmonized NAIP variant performs reasonably well but introduces more spectral error and slightly higher hallucination (SAD: 1.5671 and HA: 0.4705). The WorldStrat-trained model appears perceptually blurrier and less structurally aligned, despite having similar reflectance L1, and its SAD score (2.10) and omission space distance (0.5530) indicate weaker spectral fidelity. The likely explanation is that there is a heavy focus on areas with vegetation in the NAIP dataset, whereas the WorldStrat dataset has a slight urban area tilt.

Conversely, Fig. 6 presents an urban scene with buildings, where the performance trend shifts. Here, the WorldStrat-trained model produces the most visually convincing output with edges being sharp, and structural details like roof boundaries and roads being clearly resolved. Despite having a slightly higher spectral error (SAD: 1.8039), its visual quality surpasses that of both NAIP-trained models. The harmonized NAIP model underperforms in this case, with noticeable blurring and misalignment around fine structures, and a higher hallucination score (0.4900).

The SR images in Figs. 5 and 6 are histogram-matched with the HR ground-truth image to obtain SRharm, which is the harmonized SR output. This additional SR output is used

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:12:58 UTC from IEEE Xplore. Restrictions apply.



Fig. 6. Qualitative comparison of RGB SR results (urban scene). As in Fig. 5, rows represent different training sources. In contrast to the vegetated example, the WorldStrat-trained model produces sharper and more realistic structures in buildings and roads, suggesting stronger generalization in urban domains despite slightly higher spectral error.

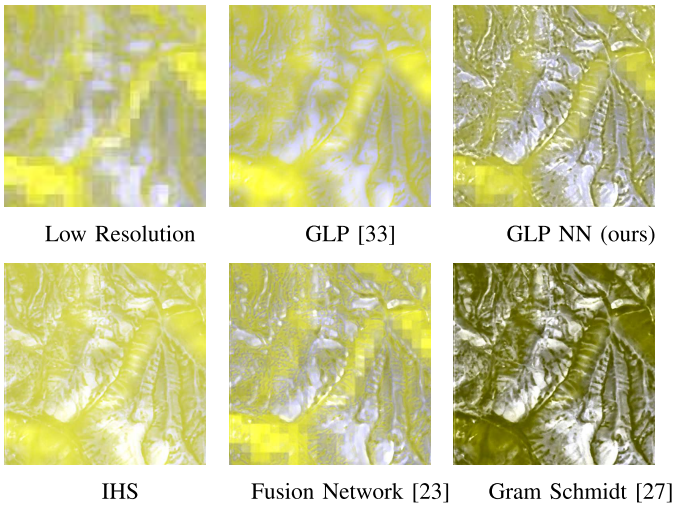


Fig. 7. Fusion output comparisons across 60 m atmospheric bands. The figure shows the LR input, GLP baseline method [31], our GLP-inspired NN, IHS transform, and fusion network baseline [21] and GS [25] method.

as a diagnostic tool: a minimal visual change post-matching indicates high intrinsic spectral fidelity.

2) Fusion Module: Fig. 7 evaluates the proposed learnable fusion model. Here, we only use the harmonized NAIP-based SR model to create the RGB super-resolved image that is further used for fusion. The columns represent the LR input, GLP baseline method [31], our GLP-inspired NN, IHS transform, and fusion network baseline [21] and the Gram–Schmidt (GS) [25] method applied to 60-m bands and visualized in RGB, where the red and green channels use the water vapor band (B10) and the blue channel uses the coastal aerosol band (B1). Compared to the baselines, which often distort the structure and exaggerate spectral contrast, our GLP-inspired fusion network output is visually more consistent with the LR signal while enhancing spatial detail.

G. Quantitative Evaluation

To obtain the average rank, each model is ranked individually for every metric: for metrics where lower values indicate better performance (reflectance, spectral, spatial, hallucination, and omission), ranks are assigned from lowest (rank 1) to highest (rank 8); for metrics where higher values are better (synthesis and improvement), ranks are assigned from highest (rank 1) to lowest (rank 8). The average rank for each model is then computed as the arithmetic mean of its ranks across all seven metrics, providing a single aggregated measure of overall performance.

1) Performance on Normalized Difference Metrics: The results in show that the proposed diffusion model, trained

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:12:58 UTC from IEEE Xplore. Restrictions apply.

TABLE I

NORMALIZED DIFFERENCE-BASED DISTANCE CALCULATION (BOLD = BEST, UNDERLINED = SECOND BEST; ↓ LOWER IS BETTER, AND ↑ HIGHER IS BETTER). "AVG. RANK" SUMMARIZES EACH MODEL'S MEAN RANK ACROSS ALL SEVEN METRICS

Model	Reflectance ↓	Spectral ↓	Spatial ↓	Synthesis ↑	Hallucination ↓	Omission ↓	Improvement ↑	Avg. Rank
ldm_baseline	0.0505 ± 0.0161	9.6923 ± 2.1742	0.0715 ± 0.0679	**0.0285 ± 0.0307**	0.6067 ± 0.2172	0.3088 ± 0.1786	0.0845 ± 0.0428	5.57
opensrmodel	0.0031 ± 0.0018	1.2632 ± 0.5878	0.0114 ± 0.0111	0.0068 ± 0.0044	0.3431 ± 0.0738	0.4593 ± 0.0781	**0.1976 ± 0.0328**	4.00
satlas	0.0489 ± 0.0086	12.1231 ± 3.1529	0.2742 ± 0.0748	0.0227 ± 0.0107	0.8004 ± 0.0641	**0.1073 ± 0.0393**	0.0923 ± 0.0266	5.71
sr4rs	0.0396 ± 0.0198	3.4044 ± 1.6882	1.0037 ± 0.1520	0.0177 ± 0.0083	0.7274 ± 0.0840	0.1637 ± 0.0572	0.1089 ± 0.0292	5.42
superimage	0.0029 ± 0.0009	1.5672 ± 1.0692	0.0132 ± 0.1131	0.0046 ± 0.0027	**0.2026 ± 0.0692**	0.6288 ± 0.0754	0.1686 ± 0.0302	5.00
DiffFuSR (NAIP Harm)	**0.0024 ± 0.0009**	**1.1103 ± 0.6197**	0.0052 ± 0.0092	0.0083 ± 0.0043	0.4580 ± 0.1020	0.3543 ± 0.0836	0.1876 ± 0.0344	**2.85**
DiffFuSR (NAIP Unharm)	0.0038 ± 0.0008	1.2320 ± 0.7256	0.0053 ± 0.0093	0.0086 ± 0.0042	0.4927 ± 0.1095	0.3373 ± 0.0870	0.1701 ± 0.0351	3.857
DiffFuSR (Worldstrat)	0.0026 ± 0.0009	1.3277 ± 0.7205	**0.0032 ± 0.0074**	0.0072 ± 0.0037	0.3915 ± 0.1209	0.4248 ± 0.1058	0.1837 ± 0.0409	3.57

TABLE II

CLIP DISTANCE (BOLD = BEST, UNDERLINED = SECOND BEST, ↓ LOWER IS BETTER, AND ↑ HIGHER IS BETTER). "AVG. RANK" SUMMARIZES EACH MODEL'S MEAN RANK ACROSS ALL SEVEN METRICS

Model	Reflectance ↓	Spectral ↓	Spatial ↓	Synthesis ↑	Hallucination ↓	Omission ↓	Improvement ↑	Avg. Rank
ldm_baseline	0.1239 ± 0.0405	12.8441 ± 2.7508	0.0717 ± 0.0683	0.0409 ± 0.0290	0.5963 ± 0.3055	**0.2327 ± 0.2238**	0.1710 ± 0.1435	5.57
opensrmodel	0.0076 ± 0.0046	1.9739 ± 1.0507	0.0118 ± 0.0108	0.0194 ± 0.0120	0.1052 ± 0.0590	0.6927 ± 0.1343	0.2021 ± 0.0854	5.00
satlas	0.1197 ± 0.0233	15.1521 ± 2.9876	0.2766 ± 0.0741	**0.0648 ± 0.0302**	0.6996 ± 0.2058	**0.0872 ± 0.0947**	0.2132 ± 0.1393	5.14
sr4rs	0.0979 ± 0.0509	22.4905 ± 2.1168	1.0099 ± 0.0439	0.0509 ± 0.0237	0.3099 ± 0.1704	0.3486 ± 0.1753	0.3415 ± 0.1042	5.28
superimage	**0.0068 ± 0.0016**	1.8977 ± 1.1053	**0.0004 ± 0.0032**	0.0130 ± 0.0073	**0.0610 ± 0.0305**	0.8524 ± 0.0586	0.0866 ± 0.0395	4.28
DiffFuSR (NAIP Harm)	0.0069 ± 0.0032	**1.6739 ± 0.8602**	0.0061 ± 0.0098	0.0238 ± 0.0120	0.2883 ± 0.2333	0.3358 ± 0.2214	0.3759 ± 0.1444	**3.00**
DiffFuSR (NAIP Unharm)	0.0109 ± 0.0030	2.0734 ± 1.0205	0.0061 ± 0.0096	0.0245 ± 0.0118	0.3272 ± 0.2400	0.2929 ± 0.2169	**0.3799 ± 0.1475**	3.92
DiffFuSR (Worldstrat)	0.0075 ± 0.0031	1.7327 ± 0.8545	0.0042 ± 0.0086	0.0205 ± 0.0105	0.2612 ± 0.2544	0.4646 ± 0.2773	0.2742 ± 0.1426	3.71

TABLE III

LPIPS DISTANCE (BOLD = BEST, UNDERLINED = SECOND BEST; ↓ LOWER IS BETTER AND ↑ HIGHER IS BETTER). "AVG. RANK" SUMMARIZES EACH MODEL'S MEAN RANK ACROSS ALL SEVEN METRICS

Model	Reflectance ↓	Spectral ↓	Spatial ↓	Synthesis ↑	Hallucination ↓	Omission ↓	Improvement ↑	Avg. Rank
ldm_baseline	0.1239 ± 0.0405	12.8441 ± 2.7508	0.0717 ± 0.0683	0.0409 ± 0.0290	0.4558 ± 0.2932	0.3558 ± 0.2518	0.1884 ± 0.1232	6.00
opensrmodel	0.0076 ± 0.0046	1.9739 ± 1.0507	0.0118 ± 0.0108	0.0194 ± 0.0120	0.0642 ± 0.0271	0.6690 ± 0.1291	0.2668 ± 0.1071	5.00
satlas	0.1197 ± 0.0233	15.1521 ± 2.9876	0.2766 ± 0.0741	**0.0648 ± 0.0302**	0.5999 ± 0.2182	**0.0588 ± 0.0552**	0.3413 ± 0.1858	5.00
sr4rs	0.0979 ± 0.0509	22.4905 ± 2.1168	1.0099 ± 0.0439	0.0509 ± 0.0237	0.3417 ± 0.1833	0.1924 ± 0.1402	0.4659 ± 0.1448	5.00
superimage	**0.0068 ± 0.0016**	1.8977 ± 1.1053	**0.0004 ± 0.0032**	0.0130 ± 0.0073	**0.0357 ± 0.0200**	0.8844 ± 0.0391	0.0800 ± 0.0301	4.28
DiffFuSR (NAIP Harm)	0.0069 ± 0.0031	**1.6738 ± 0.8602**	0.0060 ± 0.0097	0.0237 ± 0.0120	0.1149 ± 0.1020	0.3466 ± 0.1493	0.5383 ± 0.1240	**3.00**
DiffFuSR (NAIP Unharm)	0.0109 ± 0.0030	2.0734 ± 1.0205	0.0061 ± 0.0096	0.0245 ± 0.0118	0.1434 ± 0.1158	0.2846 ± 0.1508	**0.5720 ± 0.1242**	3.85
DiffFuSR (Worldstrat)	0.0075 ± 0.0031	1.7327 ± 0.8545	0.0042 ± 0.0086	0.0205 ± 0.0105	0.0752 ± 0.0682	0.6216 ± 0.1214	0.3032 ± 0.0878	3.85

TABLE IV

ABLATION METRICS ACROSS LPIPS, NORMALIZED DIFFERENCE, AND CLIP. (↓ LOWER IS BETTER; ↑ HIGHER IS BETTER). THE SELECTED MODELS FOR COMPARISON ARE TRAINED ON NAIP DATA USING THREE SETTINGS, I.E., HARMONIZED DATA WITH ANISOTROPIC BLIND KERNEL (HARM), UNHARMONIZED DATA WITHOUT HARMONIZATION, AND HARMONIZED DATA WITH A FIXED KERNEL

	LPIPS			Normalized Difference			CLIP		
Metric	Harm	Unharm	Harm (Fix Kernel)	Harm	Unharm	Harm (Fix Kernel)	Harm	Unharm	Harm (Fix Kernel)
Reflectance ↓	0.0069 ± 0.0031	0.0109 ± 0.0030	0.0329 ± 0.0187	0.0024 ± 0.0009	0.0038 ± 0.0008	0.0122 ± 0.0067	0.0069 ± 0.0032	0.0109 ± 0.0030	0.0329 ± 0.0187
Spectral ↓	1.6738 ± 0.8602	2.0734 ± 1.0205	7.3211 ± 3.1621	1.1103 ± 0.6197	1.2320 ± 0.7256	1.5652 ± 0.9009	1.6739 ± 0.8602	2.0734 ± 1.0205	7.3211 ± 3.1621
Spatial ↓	0.0060 ± 0.0097	0.0061 ± 0.0096	0.0132 ± 0.0123	0.0052 ± 0.0092	0.0053 ± 0.0093	0.0130 ± 0.0118	0.0061 ± 0.0098	0.0061 ± 0.0096	0.0132 ± 0.0123
Synthesis ↑	0.0237 ± 0.0120	0.0245 ± 0.0118	0.0570 ± 0.0325	0.0083 ± 0.0043	0.0086 ± 0.0042	0.0187 ± 0.0101	0.0238 ± 0.0120	0.0245 ± 0.0118	0.0570 ± 0.0325
Hallucination ↓	0.1149 ± 0.1020	0.1434 ± 0.1158	0.5222 ± 0.1747	0.4580 ± 0.1020	0.4927 ± 0.1095	0.7463 ± 0.0521	0.2883 ± 0.2333	0.3272 ± 0.2400	0.3810 ± 0.1223
Omission ↓	0.3466 ± 0.1493	0.2846 ± 0.1508	0.0508 ± 0.0368	0.3543 ± 0.0836	0.3373 ± 0.0870	0.1376 ± 0.0373	0.3358 ± 0.2214	0.2929 ± 0.2169	0.1003 ± 0.0289
Improvement ↑	0.5383 ± 0.1240	0.5720 ± 0.1242	0.4269 ± 0.1611	0.1876 ± 0.0344	0.1701 ± 0.0351	0.1160 ± 0.0207	0.3759 ± 0.1444	0.3799 ± 0.1475	0.5186 ± 0.1202

on harmonized NAIP data, achieves best-in-class performance in multiple key metrics, including average rank (Table I). Specifically, it achieves the lowest reflectance error (0.0024), highest spectral fidelity (1.1103), and the second-best spatial alignment (0.0052). Its synthesis score (0.0083) and omission (0.3543) are high among real competitors, while hallucination errors remain moderate compared to baselines.

2) CLIP Distance Analysis: Table II further confirms the robustness of DiffFuSR trained with NAIP harmonized data. Reflectance performance is essentially tied with Superimage, and our model delivers the best spectral score (1.6739). Its spatial error (0.0061) is among the lowest, while the improvement metric peaks at 0.3759, indicating an enhanced ability to reconstruct semantically meaningful content.

3) Insights From LPIPS Metric: LPIPS-based analysis shows that the proposed method not only achieves strong perceptual fidelity but also minimizes oversynthesis (Table III). It delivers a moderate hallucination score (0.1149), confirming that our model avoids generating too many spurious details. Additionally, it achieves the best improvement score (0.5383), reaffirming its effectiveness in reconstructing rich, high-frequency features from LR imagery.

4) Fusion-Based Enhancement of Multispectral Bands: To super-resolve all 12 Sentinel-2 bands, we combine RGB upsampling with learned fusion modeltrained to upscale 10-, 20-, and 60-m bands to a unified 2.5-m resolution. Table IV presents a comparison with classical baselines and the FusionNet baseline [21]. Since HR ground truth is not available for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:12:58 UTC from IEEE Xplore. Restrictions apply.

TABLE V

PERFORMANCE OF THE WORLDSTRAT-TRAINED SR MODEL COUPLED WITH OUR PROPOSED GLP-NN FUSION MODEL, EVALUATED ON BANDS AT 10-, 20-, AND 60-M RESOLUTION. THE BASELINES INCLUDE GLP [31], GS [25], AND FUSIONNET [21], BOLD INDICATES BEST AND UNDERLINED INDICATES SECOND BEST

Resolution	Method	PSNR	SSIM	MSE	R^2	NCC	ERGAS
10m (Ratio 4)	GLP	27.95	0.852	12.08	0.977	0.997	2.766
	GLP NN	**29.57**	**0.904**	**8.56**	**0.984**	**0.998**	**2.260**
	IHS	27.90	0.870	15.82	0.972	0.996	2.583
	GS	16.75	0.678	188.99	0.717	0.976	10.141
	PCA	17.53	0.118	254.67	0.552	0.941	10.627
	Fusion Net	29.48	0.898	8.62	0.984	0.998	2.275
20m (Ratio 8)	GLP	**24.39**	**0.836**	**15.51**	**0.952**	**0.998**	**0.789**
	GLP NN	22.01	0.757	26.26	0.916	0.997	1.030
	IHS	17.92	0.571	84.89	0.782	0.990	1.820
	GS	4.38	0.442	1584.73	-3.885	0.908	7.772
	PCA	15.09	0.195	257.04	0.295	0.970	2.948
	Fusion Net	21.10	0.767	32.86	0.904	0.997	1.288
60m (Ratio 24)	GLP	**27.92**	**0.957**	**3.19**	**0.997**	**0.999**	**0.188**
	GLP NN	23.65	0.909	5.05	0.995	0.999	0.322
	IHS	22.63	0.828	21.25	0.982	0.997	0.312
	GS	7.42	0.515	847.22	0.310	0.937	1.937
	PCA	17.79	0.473	86.33	0.926	0.987	0.689
	Fusion Net	23.16	0.918	7.21	0.992	0.999	0.380



Fig. 8. Visual comparison of ablations of our SR method on OpenSR-test [15] test image. Among all methods, DiffFuSR with NAIP harmonized produces the sharpest reconstruction with fewer artifacts, using a fixed kernel instead of blind kernel severly degrades the results.

all bands, performance must be assessed indirectly. To this end, we downsample the super-resolved outputs back to the native Sentinel-2 resolution and compare them there. Overall, our approach outperforms all classical baselines except for the GLP method [31], which remains highly competitive, particularly on the 20- and 60-m bands (see Table IV).

H. Ablations

Our experiments show that harmonization provides a consistent, albeit modest, improvement in performance.

A more critical factor is the choice of the degradation kernel. Instead of fixing the Gaussian blur kernel to a single value (e.g., $\sigma = 3.0$), we replace it with a *blind kernel* sampled across a range ($\sigma \in [2, 4]$). This change produces a dramatic improvement; with a fixed kernel, the model performance degrades sharply, while the blind kernel yields much more robust results. The importance of this choice is evident both quantitatively (Table IV) and qualitatively (see Fig. 8). Without the blind kernel, the entire pipeline nearly collapses, highlighting its central role in the training setup.

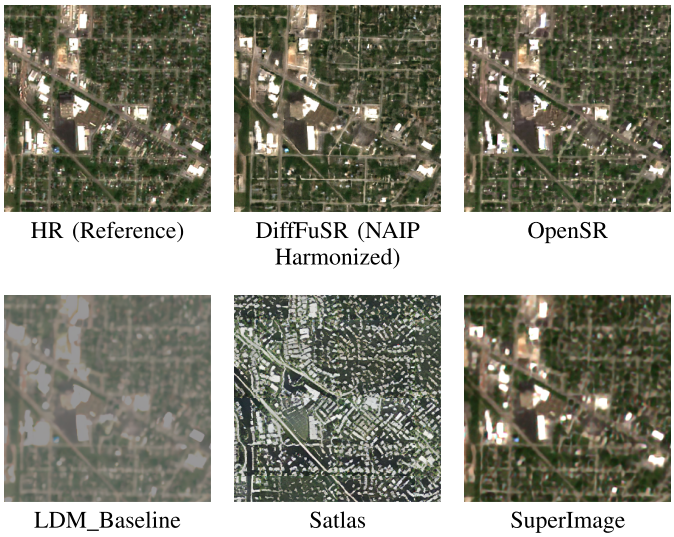


Fig. 9. Visual comparison of SR methods on an OpenSR-test [15] image. The top row highlights the strongest models: HR reference, our model trained on harmonized NAIP, and OpenSR. The bottom row shows the remaining baselines. Among all methods, DiffFuSR produces the sharpest reconstruction with fewer artifacts, while OpenSR remains a strong baseline. LDM_Baseline and Satlas show degraded outputs, and SuperImage tends to oversmooth fine structures.

V. DISCUSSION

We note that the qualitative results need to be paired with quantitative results to accurately gauge the true performance.Looking at one of the columns of results in quantitative tables can be misleading, e.g., Satlas and LDM_Baseline seem to be doing the best in terms of Omission metric in Table III. However, when looking at the rest of the values and also the qualitative results in Fig. 9, the SR outputs contain severe artifacts.

The strong performance of the harmonized NAIP model across all metric categories (Tables I–III) highlights the critical role of spectral harmonization. By aligning the source domain (NAIP) with the target (Sentinel-2), the model achieves superior reflectance and spectral fidelity, indicating it learns a more accurate representation of Sentinel-2's characteristics. Qualitatively, particularly the harmonized NAIP variant, produced sharper reconstructions with fewer artifacts compared to strong baselines like OpenSR, and markedly better results than LDM_Baseline, Satlas, or Superimage, which exhibited issues such as instability, poor spectral integrity, or oversmoothing, respectively (Fig. 9).

Traditional nongenerative SR approaches often optimize for pixel-wise fidelity, which can result in perceptually subdued images. While some modern regression-based models like Superimage offer good reflectance fidelity, our generative approach aims for, and achieves, a compelling balance of perceptual realism and quantitative accuracy. In the domain of generative models, GAN-based approaches like Satlas showed a tendency for poorer spectral integrity despite visual sharpness. Our diffusion model, benefiting from robust training strategies, including blind degradation modeling, provides a more controlled generation process.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:12:58 UTC from IEEE Xplore. Restrictions apply.

However, the superior performance of the GLP baseline when matching LR images does not translate into qualitative results. As shown in Fig. 7, the GLP baseline produces very blurry outputs, whereas our proposed method generates sharp results with well-preserved details while remaining the most accurate in terms of quantitative metrics.

A critical factor in the success of the RGB SR stage was the spectral harmonization and degradation modeling applied to the proxy HR datasets (NAIP and WorldStrat) to simulate Sentinel-2 characteristics. The blind degradation modeling, facilitated by a contrastive degradation encoder, is a significant contribution, enabling the SR model to adapt to unseen degradation conditions in real Sentinel-2 imagery without explicit kernel information.

This suggests that while harmonization is effective, the inherent spatial and radiometric properties of the proxy data can still subtly influence model specialization, highlighting the value of diverse training datasets for developing globally applicable models.

The learned fusion network, trained using only native Sentinel-2 data via a Wald protocol simulation, proved highly effective. Its ability to utilize the 2.5-m super-resolved RGB output from the diffusion model and to upscale the 10-, 20-, and 60-m bands while preserving spectral integrity is a core achievement. The scale-aware design, with dedicated fusion modules and tailored inputs for each Sentinel-2 resolution group (10-, 20-, and 60-m bands), contributes to the robustness and high fidelity of the final super-resolved Sentinel-2 12-band product. This modularity, both within the fusion stage and in the overall two-stage pipeline, offers practical benefits for future refinement and adaptation. There is room for improvement in the design choices in the fusion module architecture, to match the quantitative performance of the GLP baseline, which we leave for future work.

A. Implications of the Evaluation Strategy

A key strength of this study's SR evaluation is the use of the OpenSR-test benchmark. By assessing performance across three different distance metrics (normalized difference, LPIPS, and CLIP), we can move beyond simple pixel error. This provides a more holistic validation, capturing physical accuracy (reflectance), perceptual realism (LPIPS), and semantic consistency (CLIP), which is critical for gauging the true utility of generative models.

It is crucial to properly interpret the quantitative results for the multispectral fusion (Table V). In the absence of HR ground truth for all 12 Sentinel-2 bands, performance was assessed indirectly by downsampling the super-resolved outputs back to their native resolution and comparing them there. This "no-reference" validation, based on the Wald protocol, primarily measures fidelity to the original LR signal, not absolute accuracy at the 2.5-m target resolution. This limitation explains the apparent discrepancy noted in our results: the classical GLP method excelled in the quantitative metrics (by producing a blurry result very similar to the downsampled input) but was clearly inferior in qualitative assessment (Fig. 7), where it produces very blurry outputs. Our proposed GLP-NN, therefore, strikes a better balance, enhancing spatial detail while remaining acceptably consistent with the original spectral information.

VI. Limitations and Future Work

While DiffFuSR advances Sentinel-2 SR, several limitations remain that can be addressed in future work.

1) *Proxy Supervision and Domain Gap:* Training the RGB SR model on NAIP and WorldStrat data with statistics-based harmonization cannot fully remove sensor and illumination differences. Residual domain gaps may bias colors and textures in super-resolved outputs. *Future research could explore large-scale datasets of harmonized cross-sensor imagery, such as extended SEN2NAIP [3] variants with more temporal diversity, which could help model real-world variations more effectively.*
2) *Guidance Bias in Fusion:* Using SR-RGB as a spatial prior can transfer RGB-specific artifacts or impose structures not present in non-RGB bands (e.g., SWIR), risking subtle spectral distortions despite our safeguards. *Future work may investigate fusion models with adaptive spectral weighting to dynamically regulate RGB influence per band.*
3) *Evaluation Constraints:* In the absence of HR Sentinel-2 ground truth, we rely on OpenSR-style proxies and Wald-based no-reference validation; these may not perfectly reflect absolute radiometric fidelity in the wild. *Future evaluation can come up with suitable downstream applications that demonstrate the utility of the super-resolved bands.*

VII. Conclusion

In this work, we proposed and demonstrated DiffFuSR, a pipeline to super-resolve all 12 bands of Sentinel-2. The main contribution of the work was the combination of strengths of both classical and state-of-the-art deep learning methods, and ensuring best practices for preprocessing and harmonizing the data. The work has demonstrated that learning state-of-the-art Sentinel-2 SR models from other data modalities, e.g., NAIP, is not only possible but one of the best ways due to the lack of paired data. Converting the NAIP data to look like Sentinel-2 is the most critical factor in the pipeline. Small details like the correct kernel size and the type of blur kernel are crucial factors for an accurate degradation model to simulate Sentinel-2 images. By embedding the principles of MTF filtering and regression-based detail injection into a learnable architecture, the model inherits the interpretability and stability of traditional approaches while benefiting from the adaptability of deep learning.

Beyond the technical contributions, DiffFuSR represents a practical step toward bridging the gap between open-access data and costly HR commercial imagery. By providing a reliable method to generate a 2.5-m, 12-band Sentinel-2 product, this work can unlock new capabilities for a wide range of applications, such as detailed land cover mapping, agricultural monitoring, and urban analysis, for researchers who rely solely on the public Copernicus data stream. Ultimately, this work demonstrates that a modular pipeline, thoughtfully combining

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:12:58 UTC from IEEE Xplore. Restrictions apply.

generative models with classical principles and rigorous data handling, is key to enhancing the value of foundational EO datasets like Sentinel-2.

REFERENCES

[1] P. Kansakar and F. Hossain, "A review of applications of satellite Earth observation data for global societal benefit and stewardship of planet Earth," *Space Policy*, vol. 36, pp. 46–54, May 2016.

[2] S. Donike, C. Aybar, L. Gómez-Chova, and F. Kalaitzis, "Trustworthy super-resolution of multispectral Sentinel-2 imagery with latent diffusion," *IEEE J. Sel. Topics Appl. Earth Observat., Remote Sens.*, vol. 18, pp. 6940–6952, 2025.

[3] C. Aybar, D. Montero, J. Contreras, S. Donike, F. Kalaitzis, and L. Gómez-Chova, "SEN2NAIP: A large-scale dataset for Sentinel-2 image super-resolution," *Scientific Data*, vol. 11, no. 1, p. 1389, Dec. 2024.

[4] T. Tarasiewicz et al., "Multitemporal and multispectral data fusion for super-resolution of Sentinel-2 images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5406519.

[5] P. Kowaleczko et al., "A real-world benchmark for Sentinel-2 multi-image super-resolution," *Scientific Data*, vol. 10, no. 1, p. 644, Sep. 2023.

[6] N. Latte and P. Lejeune, "PlanetScope radiometric normalization and Sentinel-2 super-resolution (2.5 m): A straightforward spectral–spatial fusion of multi-satellite multi-sensor images using residual convolutional neural networks," *Remote Sens.*, vol. 12, no. 15, p. 2366, Jul. 2020. [Online]. Available: https://www.mdpi.com/2072-4292/12/15/2366

[7] P. Wolters, F. Bastani, and A. Kembhavi, "Zooming out on zooming in: Advancing super-resolution for remote sensing," 2023, *arXiv:2311.18082*.

[8] C. Lanaras, J. Bioucas-Dias, S. Galliani, E. Baltsavias, and K. Schindler, "Super-resolution of Sentinel-2 images: Learning a globally applicable deep neural network," *ISPRS J. Photogramm. Remote Sens.*, vol. 146, pp. 305–319, Dec. 2018.

[9] G. Vivone et al., "A critical comparison among pansharpening algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2565–2586, May 2015.

[10] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 10684–10695.

[11] C. Saharia, J. Ho, W. Chan, T. Salimans, D. J. Fleet, and M. Norouzi, "Image super-resolution via iterative refinement," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4713–4726, Apr. 2023.

[12] J. Teng et al., "Relay diffusion: Unifying diffusion process across resolutions for image synthesis," 2023, *arXiv:2309.03350,*.

[13] J. Cornebise, I. Oršolić, and F. Kalaitzis, "Open high-resolution satellite imagery: The worldstrat dataset–with application to super-resolution," in *Proc. Adv. Neural Inform. Process. Syst.*, vol. 35, 2022, pp. 25979–25991.

[14] H. Wu, N. Ni, S. Wang, and L. Zhang, "Conditional stochastic normalizing flows for blind super-resolution of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5617916.

[15] C. Aybar, D. Montero, S. Donike, F. Kalaitzis, and L. Gómez-Chova, "A comprehensive benchmark for optical remote sensing image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[16] R. Cresson, "SR4RS: A tool for super resolution of remote sensing images," *J. Open Res. Softw.*, vol. 10, no. 1, p. 1, Mar. 2022.

[17] L. Wald, *Data Fusion: Definitions and Architectures: Fusion of Images of Different Spatial Resolutions*. Paris, France: Presses des MINES, 2002.

[18] M. Irani and S. Peleg, "Improving resolution by image registration," *CVGIP, Graph. Models Image Process.*, vol. 53, no. 3, pp. 231–239, May 1991.

[19] J. Yang, J. Wright, T. S. Huang, and Y. Ma, "Image super-resolution via sparse representation," *IEEE Trans. Image Process.*, vol. 19, no. 11, pp. 2861–2873, Nov. 2010.

[20] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2016.

[21] L.-J. Deng et al., "Machine learning in pansharpening: A benchmark, from shallow to deep networks," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 3, pp. 279–315, Sep. 2022.

[22] C. Ledig et al., "Photo-realistic single image super-resolution using a generative adversarial network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2017, pp. 105–114.

[23] X. Wang et al., "ESRGAN: Enhanced super-resolution generative adversarial networks," in *Proc. Eur. Conf. Comput. Vis. Workshops (ECCVW)*, 2019, pp. 63–79.

[24] M. Sarmad, M. C. Kampffmeyer, and A.-B. Salberg, "Diffusion models with cross-modal data for super-resolution of Sentinel-2 to 2.5 meter resolution," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2024, pp. 1103–1107.

[25] M. D. Mura, G. Vivone, R. Restaino, P. Addesso, and J. Chanussot, "Global and local gram-Schmidt methods for hyperspectral pansharpening," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 37–40.

[26] J. Ho, A. N. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2024, pp. 6840–6851.

[27] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Interv. (MICCAI)*, Munich, Germany, 2015, pp. 234–241.

[28] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.

[29] W. Shi et al., "Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1874–1883.

[30] L. Wang et al., "Unsupervised degradation representation learning for blind super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10576–10585.

[31] G. Vivone, R. Restaino, and J. Chanussot, "Full scale regression-based injection coefficients for panchromatic sharpening," *IEEE Trans. Image Process.*, vol. 27, no. 7, pp. 3418–3431, Jul. 2018.

[32] B. Aiazzi, L. Alparone, S. Baronti, and A. Garzelli, "Context-driven fusion of high spatial and spectral resolution images based on oversampled multiresolution analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 10, pp. 2300–2312, Jan. 2002.

[33] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 586–595.

[34] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.

[35] Y. Blau and T. Michaeli, "The perception-distortion tradeoff," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6228–6237.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:12:58 UTC from IEEE Xplore. Restrictions apply.